18. Implement Image Cropping, Copying and Pasting to Select a Region of Interest (ROI) Using OpenCV

CODE:

```
import cv2
import matplotlib.pyplot as plt

# Read the source image
img = cv2.imread(r"C:\Users\VIPIN BABURAJ\OneDrive\Documents\computer
vision\tiger.jpg")

if img is None:
    print("Error: Image not found!")
else:
    # Get image dimensions
    height, width = img.shape[:2]

    # Select ROI size based on image dimensions
    roi_h = min(150, height)
    roi_w = min(150, width)

    # Crop the region of interest
    roi = img[0:roi_h, 0:roi_w]

    # Copy the selected region
    copied_roi = roi.copy()

    # Create a copy of the original image
    img_paste = img.copy()

    # Select a valid destination position
    x = width - roi_w
    y = height - roi_h

    # Paste the copied ROI
    img_paste[y:y+roi_h, x:x+roi_w] = copied_roi

    # Display images
    plt.figure(figsize=(12, 5))

    plt.subplot(1, 3, 1)
    plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
    plt.title('Original Image')
    plt.axis('off')

    plt.subplot(1, 3, 2)
    plt.imshow(cv2.cvtColor(copied_roi, cv2.COLOR_BGR2RGB))
    plt.title('Cropped ROI')
```

```
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(cv2.cvtColor(img_paste, cv2.COLOR_BGR2RGB))
plt.title('Image After Pasting')
plt.axis('off')

plt.tight_layout()
plt.show()

# Save the final image
cv2.imwrite('pasted_image.jpg', img_paste)

print("ROI cropped, copied, and pasted successfully!")
```

